

Elevator System Wiring

Access-Control Elevator - ESP32 - firmware build 260925-1

Power & Ground

From	To	Notes
ESP32 GND	Common GND rail	All modules share this ground
ESP32 3V3	3.3V rail	OLED and floor-switch pull-ups
External 5V supply (+)	5V rail	Servos and L298N logic
External 5V supply (-)	Common GND rail	
Motor supply (+)	L298N +12V	
Motor supply (-)	Common GND rail	

OLED Wiring (SSD1306 128x64, I2C 0x3C / 0x3D)

From	To	Notes
OLED VCC	3.3V rail	
OLED GND	Common GND rail	
OLED SDA	ESP32 GPIO 21	I2C data
OLED SCL	ESP32 GPIO 22	I2C clock

Inside Floor Buttons Wiring

From	To	Notes
Floor 1 button leg A	ESP32 GPIO 32	Internal pull-up, LOW = pressed
Floor 1 button leg B	Common GND rail	
Floor 2 button leg A	ESP32 GPIO 33	Internal pull-up, LOW = pressed
Floor 2 button leg B	Common GND rail	
Floor 3 button leg A	ESP32 GPIO 25	Internal pull-up, LOW = pressed
Floor 3 button leg B	Common GND rail	

Door & Emergency Buttons Wiring

From	To	Notes
Door button leg A	ESP32 GPIO 4	Internal pull-up, LOW = pressed
Door button leg B	Common GND rail	
Emergency button leg A	ESP32 GPIO 16	Internal pull-up, LOW = pressed
Emergency button leg B	Common GND rail	

Hall Call Buttons Wiring

From	To	Notes
1st Floor UP leg A	ESP32 GPIO 13	Internal pull-up, LOW = pressed
1st Floor UP leg B	Common GND rail	
2nd Floor UP leg A	ESP32 GPIO 5	Internal pull-up, LOW = pressed; don't hold during boot
2nd Floor UP leg B	Common GND rail	
2nd Floor DOWN leg A	ESP32 GPIO 17	Internal pull-up, LOW = pressed
2nd Floor DOWN leg B	Common GND rail	
3rd Floor DOWN leg A	ESP32 GPIO 15	Internal pull-up, LOW = pressed; don't hold during boot
3rd Floor DOWN leg B	Common GND rail	

Floor Microswitches Wiring

From	To	Notes
Floor 1 switch NO	ESP32 GPIO 34	LOW = car at floor
Floor 1 switch COM	Common GND rail	
GPIO 34	10kΩ to 3.3V rail	Required pull-up (no internal)
Floor 2 switch NO	ESP32 GPIO 35	LOW = car at floor
Floor 2 switch COM	Common GND rail	
GPIO 35	10kΩ to 3.3V rail	Required pull-up (no internal)
Floor 3 switch NO	ESP32 GPIO 39 (VN)	LOW = car at floor
Floor 3 switch COM	Common GND rail	
GPIO 39	10kΩ to 3.3V rail	Required pull-up (no internal)
Switch NC (all)	Not connected	

Lift Motor Wiring (L298N)

From	To	Notes
L298N IN1	ESP32 GPIO 26	Direction
L298N IN2	ESP32 GPIO 27	Direction
L298N ENA	ESP32 GPIO 14	PWM speed (remove ENA jumper)
L298N +5V	5V rail	Logic supply
L298N GND	Common GND rail	
L298N OUT1	Motor terminal 1	Swap OUT1/OUT2 if car moves backwards
L298N OUT2	Motor terminal 2	

Door Servos Wiring

From	To	Notes
Left servo signal	ESP32 GPIO 18	50Hz PWM
Left servo V+	5V rail	External 5V, not ESP32
Left servo GND	Common GND rail	
Right servo signal	ESP32 GPIO 19	50Hz PWM
Right servo V+	5V rail	External 5V, not ESP32
Right servo GND	Common GND rail	

Buzzer Wiring

From	To	Notes
Buzzer (+)	ESP32 GPIO 23	Passive buzzer, PWM tone
Buzzer (-)	Common GND rail	